Alessandro Pasqui

Curriculum Vitae

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AI Research Scientist with a background in statistical physics, developing **interpretable computational methods for biology**. Interested in **foundation models**, **mechanistic interpretability**, and **differentiable optimisation** to uncover biologically meaningful representations from complex biological data.

Current Position

Jan 2026 – Present

Postdoc – Explainable AI for Biological Modeling

@ CentraleSupélec – MICS Mathématiques et Informatique pour la Complexité et les Systèmes
Supervisors: **Prof. Céline Hudelot** and **Prof. Wassila Ouerdane**

Collaborators: Julien Duquesne (Scienta) and Dr. Etienne Patin (Institut Pasteur)

Designed and implemented Sparse Latent Intervention (SLI), a sparse feature selection framework for the mechanistic interpretability of **transformer-based foundation models** trained on **transcriptomic and epigenomic data**, identifying sparse sets of interpretable features that drive patient-level clinical predictions. Manuscript under review at **International Conference on Learning Representations (ICLR) 2027**.

Education

Sep 2021 – Oct 2025

PhD in Physics – AI for Biological Modeling

@ Collège de France – CIRB Centre Interdisciplinaire de Recherche en Biologie
MSCA Marie Skłodowska-Curie Actions Doctoral Fellowship
Supervisor: **Dr. Hervé Turlier**

Collaborator: Dr. Maxence Ernoult (Google DeepMind)

Thesis: **AI-based Methods for Inverse Problems in Biological Cell Systems**

VertAX: Designed and implemented a differentiable vertex model for efficient forward and inverse modeling of confluent tissues, built in JAX. Leveraged automatic differentiation and bilevel optimisation to infer optimal cellular parameters from experimental data or to reproduce target macroscopic behaviors. Manuscript: https://arxiv.org/abs/2604.06896 in preparation for **Nature Computational Science**.

ZAugNet: Designed and implemented a self-supervised generative model for z-slice augmentation in 3D bio-imaging, built in PyTorch. Combined adversarial learning and knowledge distillation to improve axial resolution in biomedical imaging. Manuscript: https://arxiv.org/abs/2503.04843 under review at **Nature Communications**.

Sep 2018 – Mar 2021

Master's Degree in Physics – Statistical and Computational Mechanics

@ Sapienza University of Rome & Italian Institute of Technology (IIT)
Supervisors: **Prof. Giancarlo Ruocco**, **Dr. Mattia Miotto**, **Dr. Giorgio Gosti**

Thesis: **High Performance Matching Algorithm Based on Orthogonal Polynomials**

Developed a computational geometric matching framework for identifying complementary regions in 2D jigsaw puzzles and 3D archaeological artifacts. Introduced orthogonal polynomial-based shape descriptors for efficient matching, applied to the IIT project on spike protein–receptor interactions in viral infections (Milanetti et al. 2020).

Sep 2014 – Jan 2018

Bachelor's Degree in Physics

@ Sapienza University of Rome
Supervisor: **Prof. Federico Ricci-Tersenghi**

Thesis: **Percolation in Models with Long-Range Interactions**

Studied, through theoretical analysis and numerical simulations, the critical point and critical exponents in percolation models involving long-range interactions.

Scientific Career

Jan 2026 – Present **Postdoc**

@ CentraleSupélec – MICS Mathématiques et Informatique pour la Complexité et les Systèmes
Supervisors: **Prof. Céline Hudelot** and **Prof. Wassila Ouerdane**
Collaborators: Julien Duquesne (Scienta) and Dr. Etienne Patin (Institut Pasteur)
Project: Developed Sparse Latent Intervention (SLI), a sparse feature selection framework for the mechanistic interpretability of transcriptomic and epigenomic foundation models.

Sep 2021 – Oct 2025 **PhD Student**

@ Collège de France – CIRB Centre Interdisciplinaire de Recherche en Biologie
Supervisor: **Dr. Hervé Turlier**
Collaborator: Dr. Maxence Ernoult (Google DeepMind)
Thesis: AI-based Methods for Inverse Problems in Biological Cell Systems

Visiting Researcher (Sep 2023 – Feb 2024)

@ Italian Institute of Technology – CLN2S Center for Life Nano- & Neuro-Science
Collaborators: **Prof. Giancarlo Ruocco**, **Dr. Mattia Miotto**, **Dr. Giorgio Gosti**
Project: Evaluated VertAX, the differentiable vertex model I developed, on real microscopy data from confluent tissues, demonstrating its applicability to experimental biological systems.

Apr 2021 – Aug 2021 **Research Engineer**

@ Collège de France – CIRB Centre Interdisciplinaire de Recherche en Biologie
Supervisor: **Dr. Hervé Turlier**
Project: Developed image-to-image translation methods based on convolutional neural networks for microscopy imaging in developmental biology, enabling bidirectional translation between transmitted-light and fluorescence microscopy.

Sep 2018 – Mar 2021 **Master Student**

@ Sapienza University of Rome & **Italian Institute of Technology**
Supervisors: **Prof. Giancarlo Ruocco**, **Dr. Mattia Miotto**, **Dr. Giorgio Gosti**
Thesis: High Performance Matching Algorithm Based on Orthogonal Polynomials

Internship 2 (Jan 2020 – Feb 2020)

@ Sapienza University of Rome – Department of Physics
Supervisor: **Dr. Francesca Colaiori**
Project: Studied the connection properties of the directed graph related to the election results for new Wikipedia administrators.

Internship 1 (Mar 2019 – Jul 2019)

@ Italian Institute of Technology – CLN2S Center for Life Nano- & Neuro-Science
Supervisors: **Prof. Giancarlo Ruocco** and **Dr. Giorgio Gosti**
Project: Studied the pairwise correlation function for neural cells during neural rosettes development through automatic segmentation algorithms based on convolutional neural networks.

Sep 2014 – Jan 2018 **Bachelor Student**

@ Sapienza University of Rome
Supervisor: **Prof. Federico Ricci-Tersenghi**
Thesis: Percolation in Models with Long-Range Interactions

Programming

JAX/Pytorch	excellent
Python	excellent
Shell Scripting	excellent
HPC/Slurm	excellent
LaTex	excellent
C/C++	advanced

Languages

English	fluent
French	fluent
Italian	fluent
Spanish	advanced

Honors and Awards

Jun 2024 **II EELISA Student Scientific Competition – 1st Prize**

Budapest University of Technology and Economics, Budapest, Hungary

Awarded first prize for outstanding research contribution in the Chemical and Bioengineering category.

Featured in multiple interviews highlighting the scientific and societal impact of the project: EELISA interview, PSL interview.

Aug 2021 **Marie Skłodowska-Curie Actions (MSCA) COFUND Fellowship**

Université PSL, Paris, France

Awarded the prestigious fellowship through the AI4theSciences Doctoral Program 2021 at Université PSL, a highly competitive European doctoral program recognizing academic excellence and research potential.

Mar 2021 **Master's Degree, Cum Laude**

Sapienza University of Rome, Rome, Italy

Awarded the Master's degree *cum laude*, recognizing academic performance in the top tier of the graduating class.

List of Publications

Sep 2026 **Which Features Matter? Sparse Latent Intervention for Prediction-Preserving Explanations**

A. Pasqui, E. Patin, J. Duquesne, W. Ouerdane

Sparse feature selection framework identifying sparse sets of interpretable features that drive the predictions of transformer-based foundation models trained on transcriptomic and epigenomic data.

Under review at **International Conference on Learning Representations (ICLR) 2027**.

Apr 2026 **VertAX: a differentiable vertex model for learning epithelial tissue mechanics**

A. Pasqui, J. M. Catacora Ocana, A. Sinha, M. Perez, F. Delbary, G. Gosti, M. Miotto, D. Caudo, M. Ernoult, H. Turlier

Preprint available on arXiv: https://arxiv.org/abs/2604.06896
Codebase: https://github.com/VirtualEmbryo/VertAX

In preparation for **Nature Computational Science**.
Citations: 6

Mar 2025 **Self-supervised z-slice augmentation for 3D bio-imaging via knowledge distillation**

A. Pasqui, S. Mahdavi, B. Vianay, A. Colin, A. McDougall, R. Dumollard, Y. A. Miroshnikova, E. Labrune, H. Turlier

Preprint available on arXiv: https://arxiv.org/abs/2503.04843
Codebase: https://github.com/VirtualEmbryo/ZAugNet

Under review at **Nature Communications**.
Citations: 1

In Preparation for Submission

2026 (expected) **AlphaMIC: synthetic microscopy image generation**

In collaboration with Dr. H. Turlier Lab.

Development of a software tool to generate realistic artificial microscopy images using cyclic generative adversarial networks.

Manuscript in preparation.

2026 (expected) **3D nuclear roughness estimation in patient-derived organoids**

In collaboration with Dr. Y.A. Miroshnikova Lab.

Analysis of 3D nuclear morphology in patient-derived colonic organoids using spherical harmonics as local shape descriptors for pathological condition classification.

Manuscript in preparation.

Scientific Programming

Sep 2026 **SLI**

Main developer and contributor of the open-source, Sparse Latent Intervention (SLI) software framework for the mechanistic interpretability of transcriptomic and epigenomic foundation models. The codebase will be released upon acceptance of the associated manuscript to the International Conference on Learning Representations (ICLR) 2027.

Apr 2026 **VertAX**

Main developer and contributor of the open-source, fully differentiable software framework implemented in JAX for forward and inverse vertex modeling and design of epithelial tissues.

To the best of my knowledge, this is the first fully differentiable implementation of a vertex model simulator. I initiated its development during my PhD research at Collège de France, and the codebase was released in April 2026. Since its release, the associated paper has already been cited by three articles authored by independent research groups worldwide, with no prior links to Collège de France, providing early evidence of external uptake.

Mar 2025 **ZAugNet**

Main developer and contributor of the open-source, self-supervised z-slice augmentation software implemented in PyTorch for 3D bioimaging using adversarial learning and knowledge distillation.

Invited Talks in International Conferences and Scientific Seminars

Nov 2026 (scheduled) **Seminar at the BioInfo Group**
Laboratoire Interdisciplinaire des Sciences du Numérique, Paris, France

Jun 2026 **Conference: Frontiers of Neuromorphic Computing II**
Max Planck Institute for the Science of Light, Erlangen, Germany

Apr 2026 **Seminar at the Laboratoire M.I.C.S.**
CentraleSupélec, Paris-Saclay, France

Feb 2024 **Seminar at the Centre Interdisciplinaire de Recherche en Biologie**
Collège de France, Paris, France

May 2022 **Seminar at the Ascidian Biocell Group**
Laboratoire de Biologie du Développement de Villefranche-sur-Mer, Villefranche-sur-Mer, France

Contributed Talks and Posters in International Conferences and Scientific Seminars

Oct 2026 (scheduled) **Data-Driven Biological Modeling (CENTURI Scientific Meeting)**
Institut d'Études Scientifiques de Cargèse, Corsica, France

Feb 2025 **Unrolling and un/self/supervised learning for inverse problems (GDR IASIS)**
Inria, Paris, France

Feb 2025 **From Soft Matter to Biophysics (Les Houches Physics School)**
École de physique des Houches, Les Houches, France

Sep 2024 **AI for Biological Physics**
Institute of Physics, London, United Kingdom

Jul 2024 **16th World Congress on Computational Mechanics (WCCM/PANACM)**
Vancouver, Canada

Jun 2024 **II EELISA Student Scientific Competition**
Budapest University of Technology and Economics, Budapest, Hungary

May 2024 **PSL Doctoral Conference – AI4theSciences**
Université Paris Dauphine-PSL, Paris, France

Jul 2023 **Artificial Intelligence in Biology and Health**
Institut Pasteur, Paris, France

Jun 2022 **PSL Doctoral Workshop – AI4theSciences**
École des Mines de Paris, Sophia Antipolis, France

Scientific Supervision

Jul 2026 - Present **Alexis Vannson (M1 student)**
CentraleSupélec / Université Paris-Saclay
Co-supervision of a Master's research project with Prof. Céline Hudelot and Prof. Wassila Ouerdane.

Jul 2026 - Present **Millan Das (M1 student)**
CentraleSupélec / Université Paris-Saclay
Co-supervision of a Master's research project with Prof. Céline Hudelot and Prof. Wassila Ouerdane.

Jul 2026 - Present **Arthur De Leusse (M1 student)**
CentraleSupélec / Université Paris-Saclay
Co-supervision of a Master's research project with Prof. Céline Hudelot and Prof. Wassila Ouerdane.

Teaching

Oct 2026 – Jan 2027 (scheduled) **Course Lecturer/Coordinator (replacing Dr. C. Bongiorno)**
@ CentraleSupélec / Université Paris-Saclay
Responsible for the course "Modélisation de la Propagation Virale" under the "Mathématiques & Data Sciences" track of the Diplôme d'Ingenieur at CentraleSupélec. Designed the curriculum and gave 7.5 hours of course-seminars covering network theory and computational projects.

May 2026 – Jun 2026 **Teaching Assistant**
@ CentraleSupélec / Université Paris-Saclay
Supervised tutorial (TD) sessions for the "Traitement de Données Massives" course in the Diplôme d'Ingenieur at CentraleSupélec, working with the course lecturer to guide students through large-scale data processing methods.

Apr 2026 – May 2026 **Teaching Assistant**
@ CentraleSupélec / Université Paris-Saclay
Supervised tutorial (TD) sessions for the "Time Series and Agent-based Modeling in Finance" course in the Diplôme d'Ingenieur at CentraleSupélec, working with the course lecturer to guide students through computational modeling exercises.

Feb 2026 – Jun 2026 **Teaching Assistant**
@ CentraleSupélec / Université Paris-Saclay
Supervised tutorial (TD) and practical (TP) sessions for the BSc in "AI, Data & Management Sciences" (CentraleSupélec–ESSEC joint degree), working with the course lecturer to guide students through Python programming exercises and data science projects.

Jan 2026 – Mar 2026 **Teaching Assistant**
@ CentraleSupélec / Université Paris-Saclay
Supervised tutorial (TD) sessions for the MSc in "Data Sciences and Business Analytics" (CentraleSupélec–ESSEC joint degree), working with the course lecturer to guide students during C++ programming exercises.

Organization of Conferences and Seminars

Sep 2023 – Aug 2025 **CIRB Seminar Series at Collège de France**
Paris, France
Member of the organizing committee for the weekly seminar series of the Biology Department at Collège de France. Responsibilities include inviting speakers, chairing sessions, and coordinating post-seminar receptions.

Professional Service

Mar 2026 **43rd International Conference on Machine Learning (ICML 2026)**
Served as a peer reviewer for papers submitted to the conference.

Apr 2024 **21st IEEE International Symposium on Biomedical Imaging (ISBI 2023)**
Served as a peer reviewer for the "Light My Cells: Bright Field to Fluorescence Imaging Challenge", organized by the French National Research Infrastructure for Biological Imaging (FBI).

Outreach and Communication

Oct 2024 **French Science Festival at Collège de France (Fête de la Science)**
Paris, France
Organized and ran a public science outreach stand, aimed at making scientific research accessible to a general audience.

Sep 2024 **European Heritage Days at Collège de France (Journées Européennes du Patrimoine)**
Paris, France
Contributed to the organization of events, helping welcome visitors and explain the historical and scientific significance of the institution.

Jun 2024 **Interviews for the II EELISA Student Scientific Competition**
Budapest, Hungary & Paris, France
Participated in interviews with EELISA and PSL about the competition.

May 2023 **Journalistic Writing Workshop with "The Conversation" Journal**
Paris, France
Training in science communication and reporting for a general audience, with a focus on communicating research activities and results.

Paris, 15 October 2026